

The Warehouse Design Challenge

Vayu Air · Session 3: Data Modeling & Warehouse Engineering

Background

Vayu Air is a fast-growing airline flying domestic and international routes out of India. Its booking system runs the business beautifully, but every analytics request goes straight to that live transactional database, and the reports are slow, hard to write, and never quite agree with each other. Revenue by route, by month, by frequent-flyer tier: each team writes its own tangle of joins and gets its own numbers.

The data team has decided to build a proper analytics warehouse so the business can answer these questions quickly and consistently, and so historical reports stay honest even as customers change over time.

The Challenge

Neha, Vayu Air's analytics engineering lead, has handed you a set of raw source tables landed from the booking system. Your job is to model them into a clean warehouse: design and build a star schema, snowflake the geography, track passenger history correctly with a slowly changing dimension, partition the large sales fact, organise everything into medallion layers, and write a data contract for the source feed. You will work in T-SQL on SQL Server.

What you'll work with

Raw, source-style tables in a `bronze` area: airports, aircraft, passengers, flights, and the bookings that record every ticket sold, plus a daily feed of passenger changes. The full data dictionary is in document 04. You will design the warehouse `dw` tables yourself.

Note Everything runs on SQL Server using SSMS. A couple of tasks ask you to read the actual execution plan, so keep "Include Actual Execution Plan" turned on as you work.